MIL-C-18096A(SHIPS)
30 June 1969
SUPERSEDING
MIL-C-18096(SHIPS)
22 June 1954
(see 6.4)

MILITARY SPECIFICATION

CHAIN, STUD LINK, ANCHOR, DIE LOCK TYPE, HIGH STRENGTH, NON-MAGNETIC

1. SCOPE

1.1 Scope. This specification covers non-magnetic high-strength die lock type chains for use on ships.

1.2 Classification. Non-magnetic die lock chains shall be of the following sizes, as specified (see 6.2).

Size - 3/4 inch
Size - 1-1/8 inches

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-Q-9858 - Quality Program Requirements.
MIL-S-17758 - Steel Bar and Steel Forging, Nickel Modified Hadfield and Nickel-Molybdenum Modified Hadfield.
MIL-P-24380 - Paint, Anchor Chain, Solvent Type, Gloss Black.

STANDARDS

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-288 - Inspection Procedure for Determining the Magnetic Permeability of Wrought Austenitic Steels.

DRAWINGS

NAVAL SHIP ENGINEERING CENTER
S2603-665947 - Size 1-1/8 Inches High Strength Die Lock Chain
S2603-667991 - Size 3/4 Inch High Strength Die Lock Chain

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd., Chicago, Illinois 60606).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. The steel used for manufacture of the chain shall conform to one of the chemical compositions in Table I. The three steels shall conform to the magnetic permeability specified in 3.10.

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Table I - Chemical compositions

Element	Alloy 1	Alloy 2	Alloy 3
	Percent	Percent	Percent
Carbon	0.70-0.90	0.70-0.90	0.12 1/
Manganese	13-15	13-15	14.5-16.0
Phosphorus	0.07 1/	0.07 1/	0.045 1/
Sulphur	0.05 1/	0.05 1/	0.030 1/
Silicon	0.50-1.00	0.50-1.00	0.30-1.00
Nickel	3.0-3.5	1.75-2.25	---
Chromium	0.50 1/	0.50 1/	17.00-19.00
Molybdenum	---	0.35-0.55	---
Nitrogen	---	---	0.35 2/

1/ Maximum
2/ Minimum

3.1.1 Tensile properties. Alloy 1 and alloy 2 shall meet all other requirements of MIL-S-17758. Alloy 3, in the as rolled condition, shall have a minimum tensile strength of 120,000 psi; a minimum yield strength of 70,000 psi, and a minimum elongation in two inches of 40 percent.

3.2 Sizes. Dimensions for size 3/4 inch chain shall be as shown on Drawing S2603-667991, and for size 1-1/8 inch, as shown on Drawing S2603-665947.

3.3 Links.

3.3.1 Form. Each link shall be composed of two forged "U" members, namely a stem member and a socketed member. The stem member shall be drop forged with tapered collars on each leg as indicated on the drawing. The socketed member of the link shall contain a recess in each leg to receive the corresponding tapered sections of the stem member, and the socketed member shall also contain the stud portion of the link. Links shall be assembled by die forging together the cold stem member and the hot socketed member.

3.3.1.1 Fit. The fit between each member and the concentricity of the stem member within the socketed member shall be as established on the drawings (see 3.2).

3.4 Length. Unless otherwise specified in the contract or order, the chain submitted to the Government shall be in 15-fathom shots. The overall length of any six consecutive links, measured from every third link, shall be within the limits specified in Table II.

3.4.1 Shortened shots. Shots shortened by the removal for test of one triplet are acceptable for shipment, providing the shot is reduced to an odd number of links.

3.5 Breaking load. When tested as specified in 4.5.2.1, the chain shall be capable of withstanding the applicable breaking load specified in Table II.

3.6 Proof load. When tested as specified in 4.5.2.2 the chain shall withstand the proof load without fracture, tendancy to open at the lock, stretching beyond the tolerances shown in Table II, or revealing any other defects which might affect the serviceability of the chain.

3.7 Impact. Chain shall be capable of withstanding the tensile impact test and compressive impact test specified in 4.5.2.3 and 4.5.2.4.

3.8 Quality.

3.8.1 Surface. Links shall be free from burrs and from excessive overlap, flaws and defects, including rough surfaces which might cause kinking of the chain in service.

3.8.2 Overall. Links shall be free from harmful defects such as cracks, seams, pipes, and laps.

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Table II - Properties and Physical test values

Size	Number of links per shot	Length of six consecutive links			Proof load	Minimum breaking load	Height (inches)		Nominal weight of 15-fathom shot
		Minimum	Nominal	Maximum			Compressive impact test	Tensile impact test	
		Inches	Inches	Inches	Pounds	Pounds	Inches	Inches	Pounds
3/4	359	19-13/32	19-1/2	19-25/32	36,000	67,500	4-25/32	3-3/16	550
1-1/8	237	29-7/64	29-1/4	29-43/64	79,500	144,900	16-3/32	10-3/4	1270

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3.9 Markings. Unless otherwise specified in the contract or order, the stud of each link shall be permanently marked, in raised letters, with the brand name, or trade name of the manufacturer and the wire diameter of the links on one side, and with the letters "USN-NM" on the other side. All markings shall be raised letters or figures of the maximum height permitted by the size of the stud, but not to exceed 3/4 inch in height by 1/8 inch in relief. The end links of each shot shall be stamped on the flat side upon submission for test with the serial numbers of the shot when so required by the Government.

3.10 Magnetic permeability. The magnetic permeability of the finished chain shall not exceed a value of 1.10.

3.11 Finish. After tumbling or grit blasting to remove scale, shots of chain shall be covered with one coat of anchor chain paint in accordance with MIL-P-24380. The finish shall be applied after it has been found that the chain otherwise complies with the requirements of this specification.

3.12 The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality control. The supplier shall provide and maintain a quality control system acceptable to the Government. The quality control system shall be in accordance with MIL-Q-9858. The supplier is in no way relieved of the final responsibility to furnish chain in accordance with the requirements specified herein.

4.2 Classification of tests. The inspection and testing of the non-magnetic die lock anchor chains shall be classified as follows:

(a) Process approval tests - Process approval tests are tests on samples submitted by the contractor to demonstrate that a uniform product can be produced by the manufacturing procedure proposed for use.
(b) Acceptance inspection tests - Acceptance inspection tests are those tests performed by the contractor on individual products or lots which have been submitted under the contract to determine conformance of the products or lots with requirements set forth in the specification prior to acceptance.

4.3 Approval of process tests. Process approval tests shall consist of the tests listed in Table III.

Table III - Process approval tests

Tests	Paragraph Reference		Number of samples
	Requirement	Method	
Breaking strength	3.5	4.5.2.1	2 triplets
Proof Test	3.6 - 3.4	4.5.2.2	2 7-link sections
Tensile Impact	3.7	4.5.2.3	2 triplets
Compressive Impact	3.7	4.5.2.4	2 single links
Macro-examination	3.8.2 3.3.1.1	4.5.2.5	single links
Magnetic Permeability	3.10	4.5.2.6	each heat of steel

4.3.1 Before actual production work is undertaken, the contractor shall demonstrate that a uniform product can be produced by the manufacturing procedure proposed for use by submitting samples for breaking load tests, proof tests, impact resistance tests, permeability tests, and macro-examination to the command or agency concerned. In addition, applicable production data and a statement of the various operations of manufacture in the sequence used together with a description of machines and apparatus used shall be submitted to the command or agency concerned. This shall include a listing of temperature control equipment used in each phase of the forging operations, and a description of furnace temperatures employed.

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4.3.2 Approval of process sampling and tests.

4.3.2.1 Set of specimens. Following the proposed manufacturing procedure, a set of specimens of the finished chain shall be prepared and submitted for test. In addition, permeability test specimens from the bar stock used shall be submitted for tests. A complete set of specimens shall consist of the following:

(a) Two triplets for breaking load tests.
(b) Two triplets for tensile impact tests.
(c) Two single links for compressive impact tests.
(d) Two seven-link sections for proof tests.
(e) One single link for macro-examination.
(f) Permeability test samples 1/2 inch diameter by 3 inches long, or 1 inch diameter by 6 inches long, machined from bar stock as received.

4.3.2.2 Whenever a set of specimens fails to conform to a specified requirement, additional samples manufactured under improved conditions may be submitted for retests. The new conditions of manufacture shall be submitted in writing.

4.3.2.3 Shipment of specimens for process approval. Unless otherwise specified in the contract or order the set of specimens for process approval (see 4.3.2.1) shall be shipped to the Commander, Boston Naval Shipyard, Boston, Massachusetts 02129, Attention: Materials Laboratory, Building 34, marked "For Tests". Permeability samples will be forwarded by the Boston Naval Shipyard to the Commanding Officer, Naval Ship Research and Development Laboratory, Annapolis, Maryland 21402, for magnetic permeability tests.

4.4 Acceptance inspection tests.

4.4.1 Process approval. Before manufacture is undertaken the Government shall determine that process approval tests have been met and that the processes used in the manufacture of the chain have been approved by the command or agency concerned.

4.4.2 Acceptance tests performed by the contractor on all production lots shall consist of the tests listed in Table IV.

Table IV - Acceptance tests

Tests	Paragraph Reference		Number of samples
	Requirement	Method	
Visual Examination	3.2 - 3.8	4.5.1	all
Breaking strength	3.5	4.5.2.1	one triplet
Proof Test	3.6 - 3.4	4.5.2.2	all
Magnetic Permeability	3.10	4.5.2.6	each heat of steel
Liquid Penetrant	3.8.2	4.5.2.7 4.5.2.7.1	all all
Macro-examination	3.3.1.1 3.8.2	4.5.2.5	single link 1/

1/ Prior to production and in any instance where forging dies are re-worked or modified.

4.4.3 Sampling. For the purpose of sampling and testing of the finish chain, a lot shall consist of not more than 10 shots of the same size chain made consecutively and from the same melt of steel.

4.4.3.1 Sampling procedures. One shot shall be selected from each lot of chain submitted for test, and from that shot one triplet shall be removed from either end for the break test.

4.4.3.2 Sampling for magnetic permeability determination. Prior to manufacture of the chain, samples of bar stock from each melt of steel shall be submitted for determination of the magnetic permeability in accordance with class 2 of MIL-STD-288.

4.4.4 Rejection.

4.4.4.1 Any shot failing to conform to the requirements of 3.2, 3.4, 3.5, 3.6 and 3.12 shall be rejected.

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4.4.4.2 If the triplet selected from the lot fails to pass the break test, the shot from which the triplet was taken shall be rejected, and the balance of the lot shall be divided into sublots of not more than 5 shots each. From each sublot, a triplet shall be taken from a shot selected at random and subjected to the break test. If the triplet from any sublot fails to pass the test, the entire lot shall be rejected.

4.4.4.3 Whenever a lot of chain fails to pass the break test, resulting in testing of sublots, then the succeeding five lots of chain shall be tested by sublotting as specified in 4.4.4.2. After five successive lots of chain have been tested without failure, the original sampling program of selecting one triplet to represent an entire lot may be resumed.

4.5 Inspection.

4.5.1 Examination. The finished chains shall be carefully examined for conformance with the applicable drawings and the requirements of this specification not covered by tests. Examination shall be conducted as specified in table V.

Table V - Classification of defects for examination of non-magnetic stud link chain.

Categories	Defects
Critical:	None defined.
Major:	
101	Chain is not the type specified die-lock, or stud link.
102	Chain is not the size specified.
103	Dimensions of the stem member are not as specified.
104	Dimensions of the socketed member are not as specified.
105	Dimensions of solid-forged link are not as specified.
106	Length of six consecutive links exceeds the limit specified.
107	Number of links in any shot is not an odd number.
108	Number of links in any shot is not as specified.
109	Link surface is not free of burrs, not free of excessive overlap.
110	Surface is rough.
111	Link stud is not identified as specified.
112	End links do not have stamped serial number of shot.
113	Chain has not been tumbled or sand blasted.
114	Chain is not coated with anchor chain paint.
115	The evaluation tests which are specified have not been passed.
116	The manufacturing process has not been approved, as required.
117	Identification tag is missing or not legible or damaged, or data are not complete.
118	Magnetic permeability exceeds the limit specified.
Minor:	None defined.

4.5.2 Tests.

4.5.2.1 Breaking strength. The breaking strength shall be determined from a triplet suitably secured in a testing machine approved by the command or agency concerned in such a way that the chain shall be free from twist, and the holding arrangement shall be such that all stresses bearing on the end links of the triplets are the same as those applied to every link tested. Where testing equipment would be endangered by a sudden break, it will be considered acceptable if the triplet is loaded to the required breaking strength and maintained at that load for 15 seconds.

4.5.2.2 Proof load. Shots or sections of shots under tests shall be tested by suitably securing them in a testing machine approved by the command or agency concerned in such a way that the chain shall be free from twist, and the holding arrangement shall be that all stresses bearing on the end links of the shot or sections under test are the same as those applied to every link tested. The chain shall be loaded to the limits specified in Table II. The overall length of any six consecutive links shall be measured from every third link after proof testing and while 10 percent of the proof load remains.

4.5.2.3 Tensile impact. Triplets for tensile impact shall be supported at one end in a vertical position and the other end shall be subjected to three blows of a 10,000 pound tup falling freely from the height specified in Table II for the applicable size.

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4.5.2.4 Compressive impact. Single links for compressive impact tests shall be supported on an anvil in an upright position and shall be subjected to three blows of a 1,640 pound tup falling freely from the height specified in Table II for the applicable size.

4.5.2.5 Macro-examination. A single link shall be cut in half on a plane parallel with both the long and short axes etched, and examined.

4.5.2.6 Magnetic permeability. Each melt of steel from which the chain is made shall be tested and have a permeability value not greater than 1.10. The test will be conducted by the Naval Ship Research and Development Laboratory Annapolis, Maryland 21402.

4.5.2.6.1 During manufacture, magnetic permeability check readings shall be taken on several random links from each work shift using a permeability indicator. Surface readings shall be taken after grinding to remove scale developed in manufacture.

4.5.2.7 Liquid penetrant inspection. The finished chain after removal of scale and prior to painting shall be inspected by the liquid penetrant method in accordance with MIL-STD-271. Links showing cracks, seams, laps or cold shuts shall be removed from the shot. If a surface discontinuity can be removed without affecting serviceability, it may be removed and the link resubmitted for penetrant inspection.

4.5.2.7.1 The stem member of each link, after removal of scale and before final assembly with the socketed member, shall be inspected by the liquid penetrant method in accordance with MIL-STD-271. Stem members showing cracks, seams, laps or cold shuts shall be rejected. If a surface discontinuity can be removed without affecting serviceability, it may be removed and the part submitted for penetrant inspection.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurement. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Bundling. Shots or lengths of chain shall be bundled or faked down and secured by a rope or cable in such a manner that each unit may be lifted.

5.2 Loading. Unless otherwise specified in the contract or order, chain shall be shipped in gondola cars and so loaded as to conform to Uniform Freight Classification Rules.

5.3 Marking.

5.3.1 For identification. Each shot or length of chain shall have securely wired to it a corrosion-resistant metal tag plainly marked with the following information:

SERIAL NUMBER_____
SIZE_____
LENGTH_____
CONTRACT NUMBER_____
CONTRACTOR_____

5.3.2 For shipment. Additional marking required by the contract or order shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. Nonmagnetic die lock chain covered by this specification is intended for use aboard non-magnetic mine sweepers as anchor chain.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Quantity.

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6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of the specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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